Certifying conditional dependence among environmental records from Bell measurements

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Abstract

Environmental records can be individually accurate while sharing the same errors. We show how retained coherence can quantify this conditional dependence without reconstructing the joint record distribution. For a balanced two-qubit state supported on span$\{|00\rangle, |11\rangle\}$, we derive the sharp compatibility boundary between a retained Clauser–Horne–Shimony–Holt (CHSH) value, the accuracy of R designated local binary records, and their conditional total correlation. The full extremum reduces to a convex problem in $R+1$ Hamming-weight probabilities, and every boundary value has a quantum realization with an unobserved remainder. Its inverse gives the least dependence compatible with Bell data and separate record calibrations. For two records it is explicit; at fixed accuracy, a fixed positive Bell excess requires dependence growing at least linearly with record count. Independent monitoring reaches the zero-dependence boundary in a Lindblad dephasing limit. A shared-memory Lindblad process instead improves record accuracy at constant Bell value, with an exact time after which dependence becomes certifiable. The result quantifies departures from conditional independence in environmental recording, under trusted state-sector and readout assumptions.

Keywords: Lindblad dynamics; quantum Darwinism; environmental records; CHSH inequality; conditional total correlation; quantum coherence

1. Introduction

When an environment monitors a qubit, distinguishable records of its pointer value constrain the coherence that the qubit can retain. If the qubit is entangled with a reference, that coherence can be monitored through a Bell measurement. A reduced Lindblad equation describes the pair's evolution [1, 2], while quantum Darwinism asks how information about the pointer becomes available in separate environmental fragments [3]. Connecting these descriptions requires knowing what each fragment records and how much additional information another record supplies.

An accurate record need not be an independent acquisition of information. In product-monitoring models, evidence accumulates across fragments through their distinguishability [4]. If several fragments instead copy one imperfect memory, each may predict the pointer accurately while repeating the same error. The dependence of interest is therefore conditional on the pointer value: correlations mediated solely by that value are expected even for independent records. Strong quantum Darwinism and spectrum broadcast structure make independence conditions explicit [5], and the failure of pairwise independence to

guarantee multipartite factorization motivates using the full conditional total correlation [6].

We ask how much of this dependence can be certified from separate record calibrations and a retained Bell signal. Our main result concerns balanced two-qubit states in $\mathrm{span}\{|00\rangle, |11\rangle\}$, a sector preserved by Bell-pair dephasing. With both branch-conditional errors of each record bounded by $\delta < 1/2$, we determine the least measured conditional total correlation compatible with the observed CHSH value. For example, two records with errors at most 0.05 are compatible with an independent-record description only up to $S \simeq 2.0358$. At an observed $S = 2.10$, the minimum conditional dependence is approximately 0.0254 nats. This minimum is attainable, so a larger universal lower bound cannot follow from the same inputs and assumptions.

The comparison with product monitoring identifies the additional question. Recent finite-fragment analyses relate readable records and residual coherence under conditional-product dynamics [7, 8]. In that setting, local readout distributions factorize; the product of their affinities bounds the retained coherence. Our result retains the same local error information while allowing arbitrary dependence between the records. It determines the entire dependence-constrained boundary and its inverse, rather than assuming the factorization whose validity is being tested. The number of variables reduces from probabilities on binary strings to $R + 1$ count probabilities without imposing exchangeability on the actual records.

Other information–correlation bounds address different outputs. CHSH violations bound quantum side information and guessing probabilities [9, 10], while accessible-information deficits bound discord and entanglement between a source and its fragments [11, 12]. These bounds do not identify the minimum conditional total correlation of R specified readouts from their individual error bounds. Witnesses based on interventions [13] or Bell correlations between environmental observers [14] also concern different measurements and targets. Here the Bell test is performed on the retained pair, and the environment supplies individual pointer calibrations. The state-sector assumption is trusted; directly measured coherence can replace the Bell input.

The mathematical method combines established tools. Convex optimization of Hellinger or Bhattacharyya affinity is developed in statistical hypothesis testing [15, 16]. For the present constraints, the essential reduction is to equalize the local errors before averaging permutations and complements, so that relative-entropy convexity applies with fixed product references. A quantum construction then realizes every extremal distribution. The resulting certificate is useful when joint record statistics are unavailable; it does not claim a general reduction in experimental effort when those statistics can be measured directly.

To show what the certificate resolves dynamically, we evaluate it during independent acquisition and during dissipative distribution of a shared memory. Reduced decoherence and environmental record structure are already known to be distinct [17], and shared errors in apparatus-mediated recording have been studied explicitly [18]. Our Lindblad examples supply trajectories for the sharp bound: one saturates the independent boundary, while the other gives an exact onset of detectable dependence at constant Bell coherence. Sections 2–3 state the calibrated inference and its quantitative consequences, Section 4 proves the reduction, and Section 5 develops these dynamical uses.

2. Bell coherence and calibrated environmental records

Let A and B be qubits, and let $F = F_1 \cdots F_R$ consist of designated disjoint environmental fragments. We first consider any joint state whose reduced AB state is supported on

$$\mathcal{H}_{\rm mc} = \mathrm{span}\{|00\rangle, |11\rangle\}, \qquad \rho_{00,00} = \rho_{11,11} = \frac{1}{2}. \tag{1}$$

The joint state need not be pure. Including its purifier and any unobserved environmental degrees of freedom in G, it can be written as

$$|\Psi\rangle_{ABFG} = \frac{|00\rangle|E_0\rangle + |11\rangle|E_1\rangle}{\sqrt{2}}, \qquad c = \langle E_0|E_1\rangle. \tag{2}$$

The conditional branch states are normalized. A local phase rotation removes the phase of c; the singular values of the resulting two-qubit correlation tensor are $1, |c|, |c|$. The optimized CHSH criterion [19, 20] therefore gives

$$S_{\max}(\rho_{AB}) = 2\sqrt{1 + |c|^2}. \tag{3}$$

Here $S_{\max}$ is the maximum over local dichotomic spin measurements. An observed CHSH value S need only satisfy $S \le S_{\max}$; its settings need not be optimal. For $S > 2$, equation (3) implies a lower bound on the branch overlap. Directly measured coherence can supply the same input, so Bell nonlocality is an operational route to the inference rather than a necessary ingredient of its classical optimization.

Each fragment F_j has a fixed binary POVM with outcome $X_j \in \{0, 1\}$. Let $P_s(x)$ be the joint distribution of $x = (x_1, \ldots, x_R)$ conditional on the outcome s of a Z_A measurement. The calibration assumption is

$$\alpha_j = P(X_j = 1 \mid s = 0) \le \delta, \qquad \beta_j = P(X_j = 0 \mid s = 1) \le \delta, \quad 0 < \delta < \frac{1}{2}. \tag{4}$$

The calibration asks whether the fixed readout guesses the Z_A outcome correctly in each branch. Both conditional errors are bounded; a bound only on their average $(\alpha_j + \beta_j)/2$ would leave a different feasible class. The records need not have equal errors or optimal measurements.

We quantify the dependence of these readouts by their conditional total correlation [21],

$$K = \frac{1}{2} \sum_{s=0}^{1} D\left(P_s \middle\| \prod_{j=1}^{R} P_s^j \right), \tag{5}$$

where P_s^j is the j-th marginal and D is classical relative entropy with natural logarithms. Thus K is measured in nats. It vanishes exactly when the readouts factorize in each branch. For two records, K is their conditional mutual information given the pointer value; for more records it also detects dependence that can be invisible in pairwise marginals. It quantifies the specified readout distributions, rather than every correlation in the quantum environment.

The first link between records and coherence follows directly from the readout POVM. If M_x is its joint effect on the environment, then

$$\begin{aligned} |c| &= \left| \sum_x \langle E_0|M_x|E_1\rangle \right| \\ &\le \sum_x \sqrt{\langle E_0|M_x|E_0\rangle \langle E_1|M_x|E_1\rangle} = B(P_0, P_1), \end{aligned} \tag{6}$$

where $B(P_0, P_1) = \sum_x \sqrt{P_0(x)P_1(x)}$ is the Bhattacharyya affinity. It equals one for identical distributions and zero for distributions with disjoint supports. Retained coherence therefore requires some overlap between the possible record outcomes in the two branches. We now maximize that overlap subject to the measured accuracy and a budget on conditional dependence.

3. Sharp bounds and the inference from Bell data

For $R \ge 2$, independent records with error δ have a binomial distribution for the number of outcomes equal to one in branch zero. This supplies the reference distribution

$$q_k = \binom{R}{k} \delta^k (1-\delta)^{R-k}, \qquad k = 0, \dots, R. \tag{7}$$

We will show that the largest compatible coherence can be found by varying a count distribution r_k, where k is the number of ones, while fixing its mean to $R\delta$. For $\kappa \ge 0$, define

$$\begin{aligned} &V_{R,\delta}(\kappa) = \max_{r_0,\dots,r_R} \sum_{k=0}^{R} \sqrt{r_k r_{R-k}} \\ &\text{subject to } r_k \ge 0, \quad \sum_k r_k = 1, \quad \sum_k k r_k = R\delta, \quad D(r\|q) \le \kappa. \end{aligned} \tag{8}$$

The objective is concave and the constraints define a compact convex set. Equation (8) is therefore a convex optimization problem in the concave-maximization convention. Its dimension grows linearly with the number of records.

Theorem 1 (sharp record–coherence bound).
Under equations (1), (4) and (5), if $K \le \kappa$, then

$$|c| \le V_{R,\delta}(\kappa), \qquad S_{\max} \le 2\sqrt{1 + V_{R,\delta}(\kappa)^2}. \tag{9}$$

For every $R \ge 2$, $0 < \delta < 1/2$ and $\kappa \ge 0$, the upper boundary is attained by a joint quantum state with R local binary records and an unobserved remainder. The attaining records have equal symmetric errors δ, and their specified readouts are also Helstrom-optimal.

Sharpness includes the physical realization, not just the optimization of classical probabilities. Within the stated class, no smaller upper bound follows from the three inputs R, δ, κ. Section 4 proves the reduction and attainment; we first spell out the bound's endpoints and its use as an inference.

3.1. Independent and fully shared errors

The two endpoints expose the physical role of the dependence budget. Write

$$b = 2\sqrt{\delta(1-\delta)}, \qquad h(\delta) = -\delta \ln \delta - (1-\delta)\ln(1-\delta). \tag{10}$$

At zero budget, relative-entropy positivity forces $r = q$, and hence

$$V_{R,\delta}(0) = b^R. \tag{11}$$

For independent readouts, the joint affinity is the product of the single-record affinities, and bounds the quantum branch overlap. At the other endpoint, the distribution $r_0 = 1-\delta,\ r_R = \delta$ makes every error shared. It gives

$$V_{R,\delta}(\kappa) = b \quad \Longleftrightarrow \quad \kappa \geq (R-1)h(\delta). \tag{12}$$

To verify necessity, marginalization to any one bit bounds the joint affinity by b. Equality requires the likelihood ratio P_1/P_0 to be constant within each value of that bit. For every bit these ratios are $\delta/(1-\delta)$ at zero and its reciprocal at one. Because they differ, a mixed-bit string cannot carry positive probability at equality. Only 0^R and 1^R remain, and the marginal conditions fix their masses. Their dependence is $(R-1)h(\delta)$.

The function $V_{R,\delta}$ is nondecreasing and concave in its budget. Concavity follows by mixing feasible distributions, using convexity of KL and concavity of the objective. Below (12), an optimizer uses its whole budget: otherwise a sufficiently small mixture with the shared-error endpoint would increase the objective while remaining feasible.

3.2. The least dependence compatible with an observation

For an observed $2 < S \leq 2\sqrt{2}$, define

$$c_{\text{obs}} = \sqrt{S^2/4 - 1}. \tag{13}$$

Since $S \leq S_{\max}$, equations (3) and (9) imply

$$\begin{aligned} K \geq K_{\min}(R, \delta, c_{\text{obs}}) := \min_r\, & D(r\|q) \\ \text{subject to } & r_k \geq 0, \quad \sum_k r_k = 1, \quad \sum_k k r_k = R\delta, \\ & \sum_k \sqrt{r_k r_{R-k}} \geq c_{\text{obs}}. \end{aligned} \tag{14}$$

The superlevel set of the concave affinity is convex. Equation (14) is consequently a convex minimization, and quantum attainment makes its lower bound sharp. It equals zero when $c_{\text{obs}} \leq b^R$. If $c_{\text{obs}} > b$, the Bell value and calibration are incompatible with the stated state sector.

The inputs to this inference are the record count, a Bell value and a common upper bound δ on the calibrated conditional errors. The sharpness claim refers to those summarized inputs; retaining unequal record-specific bounds could give a stronger conclusion. No estimation of the full joint distributions P_s is required. The relevant setting is one in which Bell monitoring and separate local calibrations are available but joint record

outcomes have not been collected. For two records that can be read jointly, estimating two 2×2 conditional tables may be simpler than performing this inference. For larger record sets, the bound avoids reconstructing their joint entropy, although we do not claim a general sample-complexity advantage. If the joint distributions are already known accurately, their direct affinity in (6) supplies at least as strong a constraint as compression to K.

3.3. An explicit two-record certificate

For $R = 2$, an exchangeable complement-symmetric pair with the required marginals has

$$P_0 = (1 - 2\delta + u, \delta - u, \delta - u, u), \qquad P_1 = P_0 \circ \text{complement}. \tag{15}$$

The relevant interval is $\delta^2 \le u \le \delta$. On this interval,

$$B_\delta(u) = 2\sqrt{u(1 - 2\delta + u)} + 2(\delta - u), \tag{16}$$

and

$$\begin{aligned} K_\delta(u) = (1 - 2\delta + u) \ln \frac{1 - 2\delta + u}{(1 - \delta)^2} \\ + 2(\delta - u) \ln \frac{\delta - u}{\delta(1 - \delta)} + u \ln \frac{u}{\delta^2}. \end{aligned} \tag{17}$$

We use $0 \ln 0 = 0$. The derivatives establish monotonicity without an optimization assumption:

$$\begin{aligned} B'_\delta(u) &= \frac{1 - 2\delta + 2u}{\sqrt{u(1 - 2\delta + u)}} - 2 > 0, \\ K'_\delta(u) &= \ln \frac{u(1 - 2\delta + u)}{(\delta - u)^2} > 0 \quad \text{for } u > \delta^2. \end{aligned} \tag{18}$$

For the first inequality, squaring the positive numerator and subtracting four times the denominator squared leaves $(1 - 2\delta)^2$. For the second, the numerator inside the logarithm exceeds its denominator exactly when $u > \delta^2$. Values below δ^2 are dominated by independence, which has both smaller dependence and greater affinity.

Solving $B_\delta(u) = c$ gives the explicit inverse

$$K_{\min}(2, \delta, c) = K_\delta\left(\frac{(c/2 - \delta)^2}{1 - c}\right), \qquad b^2 < c \le b. \tag{19}$$

For example, two records with both conditional errors at most 0.05 and an observed $S = 2.10$ require

$$K \ge K_{\min}(2, 0.05, \sqrt{2.10^2/4 - 1}) \simeq 0.02540 \text{ nats}. \tag{20}$$

Their independent-record limit is $S \le 2.03578$. The general affinity-angle bound in Appendix B gives a lower estimate of approximately 0.00454 nats for the same observation. Figure 1(a) shows the gain in certified dependence from solving the constrained problem exactly; this is an illustrative comparison with that estimate, not a claim of superiority to every prior inequality. Figure 1(b) shows how the same boundary extends to larger

record counts. Its saturation value is independent of R, but the dependence required to reach it grows with the number of records.

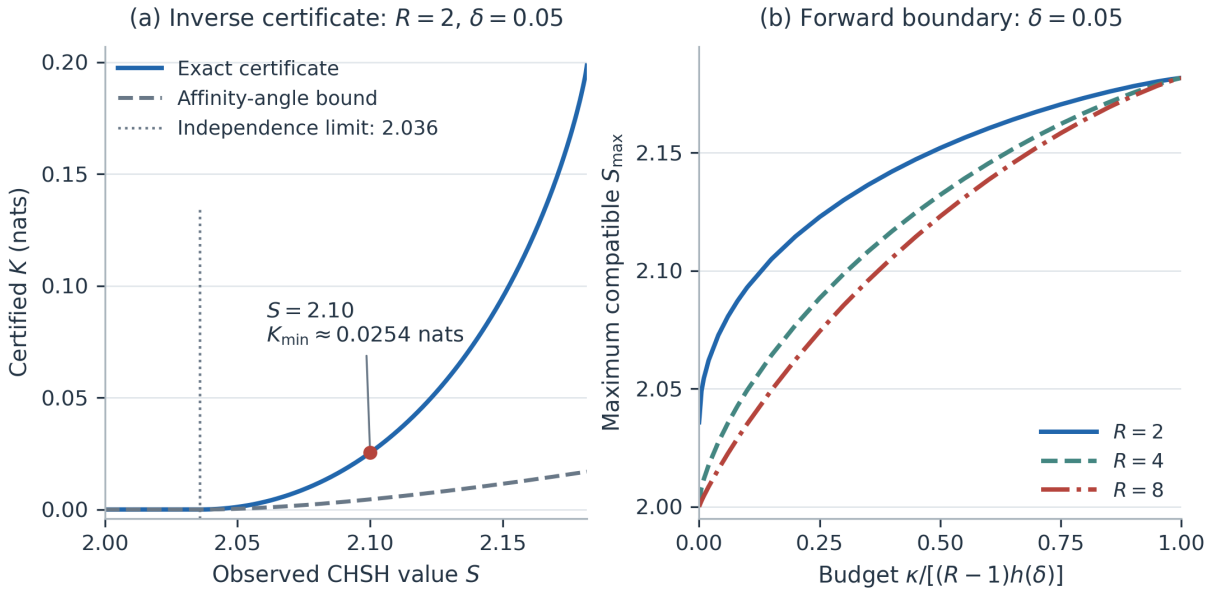


Figure 1. Sharp inference and compatibility bounds at symmetric error bound $\delta = 0.05$. (a) For two records, an observed CHSH value gives the minimum conditional total correlation K consistent with the stated sector; at $S = 2.10$ this minimum is approximately 0.0254 nats. The dashed comparison is the general affinity-angle bound in Appendix B. (b) The largest compatible CHSH value for $R = 2, 4, 8$ as dependence is allowed to increase. The normalized budget divides by $(R-1)h(\delta)$, so equal horizontal positions at different R represent different absolute budgets. The right endpoint is attained when all records share one error. Curves are calculated bounds, not experimental data.

4. Proof and dependence cost

The main difficulty is that total correlation uses a product of the distribution's own marginals. It is not generally convex when those marginals change, so averaging arbitrary record distributions would not justify the desired bound. We first process each readout to give exactly the same symmetric error. This fixes the product references without increasing dependence or decreasing affinity. Permutation and complement averaging can then reduce the extremum to the count distribution in (8). A final quantum construction is needed to show that the classical maximum is physically attainable.

4.1. Reduction to equal symmetric errors

Consider one readout with errors $\alpha, \beta \le \delta$. Apply a stochastic map that flips outcome zero with probability f_0 and outcome one with probability f_1, where

$$v = \frac{1-2\delta}{1-\alpha-\beta}, \qquad f_0 = \delta - v\alpha, \qquad f_1 = \delta - v\beta. \tag{21}$$

Because $\alpha, \beta \le \delta < 1/2$, these are valid probabilities and $f_0 + f_1 = 1 - v$. The two new errors are $f_0 + v\alpha = \delta$ and $f_1 + v\beta = \delta$. Applying these maps independently to

the records cannot increase their conditional total correlation: in each branch, relative-entropy data processing maps the joint distribution and its product of marginals to the corresponding processed distributions. The same processing cannot decrease affinity. It is therefore sufficient to prove the bound for equal symmetric errors.

This reduction also explains why heterogeneous calibrations may safely be summarized by a common upper bound δ. It gives a worst-case guarantee; it does not assert that all actual records have equal quality.

4.2. Permutation and complement symmetries

In the equal-error case, the product reference distribution in branch zero is

$$Q_0(x) = \delta^{|x|}(1-\delta)^{R-|x|}, \qquad Q_1(x) = Q_0(\bar{x}), \tag{22}$$

where $|x|$ is the Hamming weight and $\bar{x}$ is the bitwise complement. Average the pair (P_0, P_1) over all simultaneous permutations of the records. The prescribed marginals remain unchanged. Joint concavity of B cannot lower the affinity, while convexity of relative entropy with the fixed references Q_s cannot raise the average dependence cost.

Next average this pair with $(P_1 \circ \text{complement}, P_0 \circ \text{complement})$. The same inequalities apply because complementation interchanges Q_0 and Q_1. The resulting distributions are exchangeable and satisfy

$$P_0(x) = \frac{r_{|x|}}{\binom{R}{|x|}}, \qquad P_1(x) = P_0(\bar{x}). \tag{23}$$

The marginal error condition becomes $\sum_k k r_k = R\delta$. Summing by Hamming weight gives

$$B(P_0, P_1) = \sum_k \sqrt{r_k r_{R-k}}, \qquad D(P_0 \| Q_0) = D(P_1 \| Q_1) = D(r \| q). \tag{24}$$

Every original distribution has therefore been mapped to a feasible point of (8) with no smaller affinity and no larger dependence. Together with (6), this proves the upper bound. The symmetrization is an extremal argument; exchangeability is not an assumption about the experimental records.

4.3. Quantum attainment

Conversely, any feasible vector r defines the distributions in (23). On qubit registers F and G, prepare

$$|E_s\rangle = \sum_{x \in \{0,1\}^R} \sqrt{P_s(x)}\, |x\rangle_F |x\rangle_G. \tag{25}$$

Tracing out G leaves diagonal conditional states on F. Each fragment has conditional probabilities $(1-\delta, \delta)$ or their reversal, so a computational-basis readout has error δ and achieves the Helstrom optimum. Moreover,

$$\langle E_0 | E_1 \rangle = B(P_0, P_1). \tag{26}$$

Substitution in (2) attains (9). Appending an unobserved branch tag with any overlap between zero and one reduces $|c|$ without changing the readout distributions. Thus all lower coherence values compatible with the same records are realizable as well. This completes the proof.

The unobserved remainder is part of the sharpness statement. A restriction to a pure environment composed exclusively of the designated fragments would define a different extremal problem. Appendix A gives a sequential realization of (25) with memory dimension $R + 1$.

4.4. Dependence cost as the record count grows

The inverse also yields a simple conclusion about large record sets. For $0 < c \leq b$, put

$$a(c) = \frac{1 - \sqrt{1 - c^2}}{2}, \qquad I_\delta = -\ln b.$$

Then a retained overlap at least c requires

$$K \geq \max\{0,\ R\, a(c) I_\delta - h(a(c))\}. \tag{27}$$

To see this, apply randomized-tie majority decoding to the Hamming weight after the symmetry reduction. If its branch error is e, affinity data processing gives $c \leq 2\sqrt{e(1-e)}$, hence $e \geq a(c)$. Under the binomial reference, its error ϵ_R is at most e^{-RI_δ} by the exponential-moment bound for $k \geq R/2$. Binary KL data processing yields $D(r\|q) \geq D(\mathrm{Bern}(e)\|\mathrm{Bern}(\epsilon_R))$. When $a(c) > \epsilon_R$, monotonicity of binary KL in its first argument gives

$$K \geq D(\mathrm{Bern}(a(c))\|\mathrm{Bern}(\epsilon_R)) \geq a(c)\ln(1/\epsilon_R) - h(a(c)) \geq Ra(c)I_\delta - h(a(c)).$$

If $a(c) \leq \epsilon_R \leq e^{-RI_\delta}$, then $h(a(c)) \geq a(c)\ln[1/a(c)] \geq Ra(c)I_\delta$, so the displayed nonzero candidate in (27) is nonpositive. This proves (27) in both cases without assuming that the majority error e is at most $1/2$.

At fixed accuracy and fixed positive CHSH excess, $c_{\mathrm{obs}} > 0$ is fixed, so the required total correlation grows at least linearly with R. The full frontier is sharper at finite R; equation (27) identifies the scaling consequence of retaining coherence while the number of accurate records increases.

5. Lindblad dynamics of acquisition and redistribution

The state-level bound can be applied at each time of a specified dynamics. To do so, one must define the environmental fragments and their readouts alongside the reduced master equation. We give two realizations whose different trajectories explain why record accuracy cannot be interpreted from the Bell signal alone.

5.1. Independent monitoring

Prepare AB in $|\Phi^+\rangle$, and couple A to R independent streams of fresh pure ancillas. In a time interval Δt, the conditional ancilla states in stream j are chosen to have overlap

$$\langle e_0^{(j)}|e_1^{(j)}\rangle = e^{-2\gamma_j \Delta t}. \tag{28}$$

For instance, the states $\cos\vartheta_j|0\rangle \pm \sin\vartheta_j|1\rangle$, with $\cos(2\vartheta_j) = e^{-2\gamma_j\Delta t}$, are produced by a pointer-controlled rotation. After N collisions, $t = N\Delta t$, each complete stream fragment has overlap $b_j(t) = e^{-2\gamma_j t}$. The reduced channel is exactly the dephasing semigroup at these times, with continuous-time limit

$$\dot\rho_{AB} = \gamma(Z_A\rho_{AB}Z_A - \rho_{AB}), \qquad \gamma = \sum_j \gamma_j. \tag{29}$$

The streams are conditionally independent. For equal rates $\gamma_j = \gamma/R$, their equal-prior Helstrom readouts have symmetric errors

$$\delta(t) = \frac{1-\sqrt{1-e^{-4\gamma t/R}}}{2}, \qquad K(t) = 0, \tag{30}$$

and

$$S_{\max}(t) = 2\sqrt{1+e^{-4\gamma t}}. \tag{31}$$

Since $2\sqrt{\delta(t)[1-\delta(t)]} = e^{-2\gamma t/R}$, this trajectory saturates (11) at every positive finite time. Each fragment here is an entire stream up to the observation time. The construction does not assert that a finite collection of closed ancilla qubits with a bounded time-independent Hamiltonian generates an exact irreversible semigroup.

This example gives the usual interpretation of decoherence as independent information acquisition. The next model preserves a record that has already been acquired and follows its subsequent distribution.

5.2. Dissipative distribution of a shared memory

Let M be a memory qubit, and prepare

$$|\Omega\rangle_{ABM} = \frac{|00\rangle|m_0\rangle + |11\rangle|m_1\rangle}{\sqrt{2}}, \tag{32}$$

where

$$|m_0\rangle = \sqrt{1-\varepsilon}|0\rangle + \sqrt{\varepsilon}|1\rangle, \qquad |m_1\rangle = \sqrt{\varepsilon}|0\rangle + \sqrt{1-\varepsilon}|1\rangle, \quad 0<\varepsilon<\frac{1}{2}. \tag{33}$$

The memory's computational-basis value is an imperfect pointer record with error ε. It is this value that the subsequent dynamics copies. The designated record qubits initially have state $(I/2)^{\otimes R}$, independent of ABM.

On MF, use the time-independent Lindblad generator

$$\dot\rho_{ABMF} = \sum_{j=1}^{R}\sum_{s=0}^{1}\left(L_{js}\rho L_{js}^\dagger - \frac{1}{2}\{L_{js}^\dagger L_{js},\rho\}\right), \tag{34}$$

with

$$L_{j0} = \sqrt{\nu}\,|0\rangle\langle 0|_M \otimes |0\rangle\langle 1|_{F_j}, \qquad L_{j1} = \sqrt{\nu}\,|1\rangle\langle 1|_M \otimes |1\rangle\langle 0|_{F_j}. \tag{35}$$

Identity operators on the remaining systems are implicit. Conditional on the memory's basis value, each record relaxes toward that value at rate ν. The probability that a record disagrees with the memory is

$$e_t = \frac{1}{2}e^{-\nu t}.$$

After tracing out M, the record distributions are

$$\begin{aligned} P_0(x,t) &= (1-\varepsilon)e_t^{|x|}(1-e_t)^{R-|x|} \\ &\quad + \varepsilon(1-e_t)^{|x|}e_t^{R-|x|}, \qquad P_1(x,t) = P_0(\bar{x},t). \end{aligned} \tag{36}$$

The off-diagonal blocks of the memory do not contribute to these probabilities. The two diagonal blocks evolve by independent relaxation, which proves (36) even though the initial memory branches in (33) are coherent.

The local readout error and conditional total correlation are consequently

$$\delta_t = \varepsilon + (1-2\varepsilon)e_t, \qquad K_R(t) = Rh(\delta_t) - H(P_0(\cdot,t)). \tag{37}$$

Here $H(P) = -\sum_x P(x)\ln P(x)$ is the Shannon entropy in nats, with $0\ln 0 = 0$.

Trace preservation of a channel acting only on MF leaves the reduced AB state unchanged. This standard invariance supplies a controlled case in which the Bell signal alone cannot reveal the progress of recording. Equivalently, every Bell operator on AB lies in the kernel of the adjoint generator in (34), so

$$S_{\max}(t) = 2\sqrt{1+4\varepsilon(1-\varepsilon)} \quad \text{for all } t. \tag{38}$$

The record probabilities nevertheless contain the decaying modes $e^{-m\nu t}$, $m = 0, \ldots, R$. Their error decreases from $1/2$ to ε, and their conditional dependence increases from zero to $(R-1)h(\varepsilon)$. To see monotonicity, take $0 \le t_1 < t_2$ and flip each later readout independently with probability $a = (e_{t_1} - e_{t_2})/(1-2e_{t_2}) \in [0, 1/2]$. This maps its disagreement probability with the memory from e_{t_2} to e_{t_1} and recovers the earlier distribution in each branch. Relative-entropy data processing therefore gives $K_R(t_1) \le K_R(t_2)$. For every $t > 0$, the conditional covariance of two record bits is $\varepsilon(1-\varepsilon)(1-2e_t)^2 > 0$, so $K_R(t) > 0$.

The Bell measurement can remain fixed throughout this process. With $c_0 = 2\sqrt{\varepsilon(1-\varepsilon)}$, choose $A_0 = Z_A$, $A_1 = X_A$ and $B_{0,1} = (Z_B \pm c_0 X_B)/\sqrt{1+c_0^2}$. These settings attain (38) at every time, so the certification experiment does not require repeated optimization of the Bell settings.

In the long-time limit, all records copy the same noisy memory value and attain the shared-error boundary (12). The trajectory thus connects initially uninformative fragments to the shared-error endpoint under an explicit Lindblad generator. Increasing conditional dependence here occurs within a time-independent joint quantum dynamical semigroup; it does not diagnose non-Markovianity. The model concerns distribution after acquisition, and its initial Bell value differs from that of the independent-monitoring example.

5.3. When the Bell data first certify dependence

The dependence in (37) is positive immediately, but the certificate need not detect it while each record is still inaccurate. Define

$$\theta = 4\varepsilon(1-\varepsilon).$$

The independent-record hypothesis becomes incompatible with the constant Bell value (38) when

$$[4\delta_t(1-\delta_t)]^R < \theta. \tag{39}$$

Using $4\delta_t(1-\delta_t) = 1-(1-2\varepsilon)^2(1-e^{-\nu t})^2$, one obtains the exact threshold

$$t_{\mathrm{cert}} = -\frac{1}{\nu}\ln\left[1-\frac{\sqrt{1-\theta^{1/R}}}{1-2\varepsilon}\right]. \tag{40}$$

For $t > t_{\mathrm{cert}}$, equation (14) gives a strictly positive lower bound on $K_R(t)$. At $\varepsilon = 0.05$, the dimensionless thresholds νt_{cert} are approximately 1.799, 1.043 and 0.656 for $R = 2, 4, 8$. Figure 2 follows the inference for two records: as the local error falls, the Bell value eventually exceeds what independent records could permit, and the inferred dependence becomes positive. The actual dependence has already grown continuously from zero; the crossing marks the sensitivity of this certificate.

6. Conditions for using the certificate

The inference uses trusted information about the state sector and the local readouts. Its statistical implementation can therefore be separated from its deterministic bound. For repeated independent preparations, use a simultaneous lower confidence bound S_L for the Bell value and upper bounds δ_U for every branch-conditional error. When $S_L > 2$ and $0 < \delta_U < 1/2$, substitute S_L, δ_U in (14); without a Bell violation or informative calibration, these inputs give no positive certificate. Bell settings should be fixed in advance or validated on separate data.

For example, replacing the nominal values $S = 2.10, \delta = 0.05$ by conservative limits $S_L = 2.08, \delta_U = 0.055$ still gives

$$K \geq K_{\min}(2, 0.055, \sqrt{2.08^2/4-1}) \simeq 0.00723 \text{ nats}. \tag{41}$$

This example uses conservative simultaneous confidence limits, rather than substituting nominal estimates into a deterministic inequality. A concrete Hoeffding construction and its sufficient sample counts are given in the computational supplement. Appendix C treats population imbalance, leakage and an amplitude-damping family, including a natural reservoir realization for which the certificate has zero sensitivity.

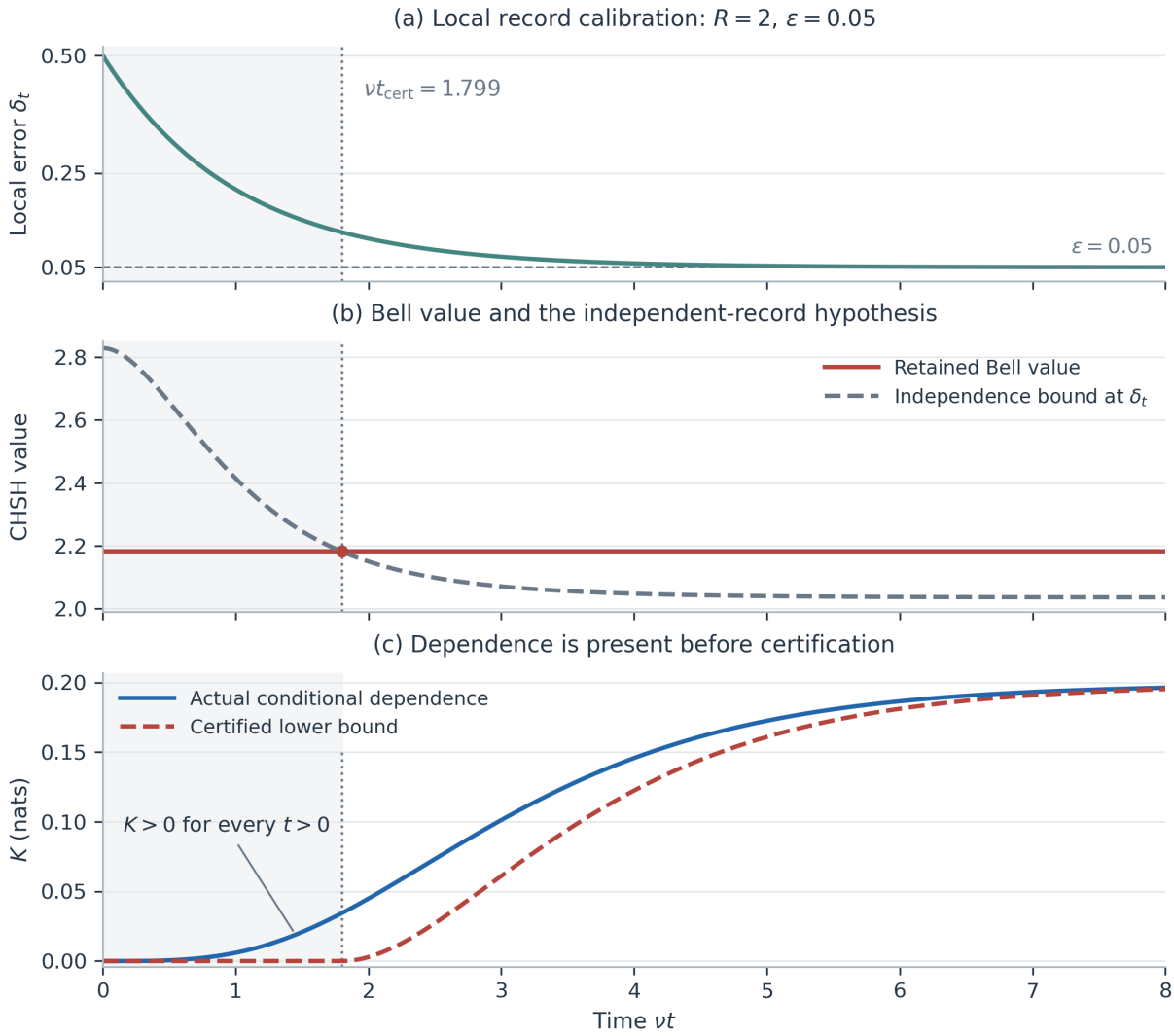


Figure 2. Applying the certificate during shared-memory Lindblad evolution, with $R = 2$ and memory error $\varepsilon = 0.05$. (a) The calibrated local readout error δ_t decreases from $1/2$ toward 0.05. (b) The pair retains $S_{\max} \simeq 2.182$, while the independence bound evaluated at that improving calibration falls below it. (c) The actual conditional total correlation is positive for all $t > 0$; its certified lower bound becomes positive after $\nu t_{\mathrm{cert}} \simeq 1.799$. Aligned vertical lines identify this certification time. All panels describe one calculated trajectory; the independence curve is a compatibility bound.

The inference also requires the recorded observable to be identified. Our calibration concerns the local pointer Z_A. Records of a collective observable, such as $Z_A Z_B$, may coexist with Bell correlations protected within its eigenspaces [22, 23]. Appendix D gives a matched local/parity example with equal initial Bell values and information curves but different subsequent Bell dynamics. It delimits the calibration assumption; it is not evidence that arbitrary accurate environmental records constrain local coherence in the same way.

7. Computational verification

The proofs establish the extremal relation; numerical calculations evaluate its boundary and independently check the quantum realizations. The computational supplement contains the complete Python source, parameter choices and diagnostics. The figures are calculated model results, not experimental data.

For Figure 1, we evaluated 81 frontier points at $R = 2, 4, 8$ and $\delta = 0.05$. The general interior solution is checked by comparing a feasible primal distribution with the variational upper bound in Appendix B; the largest overlap gap was below 8×10^{-13}. Nine further explicit $ABFG$ constructions checked quantum attainment and local Helstrom errors. The shared-memory model was tested at 21 configurations by exponentiating its sparse joint Lindblad generator, rather than evaluating only the analytic probabilities, with discrepancies below 2×10^{-15}. Another 26 density-matrix calculations checked the local/parity example. These floating-point checks support reproducibility and test the implementations; the analytic arguments carry the claims of validity and sharpness.

8. Discussion

The result makes conditional independence quantitatively testable within a specified recording sector. Separate calibrations supply a common error bound for predicting the local pointer, while the retained Bell value bounds how distinguishable the full branches can have become. Their compatibility determines a minimum dependence among the readouts. Quantum attainment ensures that this lower bound cannot be improved using only the record count, common error bound and coherence lower bound within the stated sector. The substantive gain is the finite-accuracy boundary for arbitrary conditional dependence, including an explicit two-record inverse and an $R + 1$-variable evaluation for larger record sets.

This changes the interpretation of a large set of accurate records. At fixed local accuracy, conditional independence forces the compatible coherence to decrease exponentially with record count. Retaining a fixed positive Bell excess instead requires conditional total correlation growing at least linearly with R. Many records can therefore preserve a common uncertainty even when each is useful to an observer. The shared-memory Lindblad example realizes that behavior continuously: calibration improves while the pair's state remains fixed, and the certificate detects when a conditionally independent record description becomes incompatible with the available observations.

The conclusion concerns measured dependence, not a complete criterion for quantum objectivity. The fragments and their readouts are designated; the record count is not optimized over all environment partitions. A positive certificate rejects conditional independence but does not identify a unique mechanism that generated the dependence. A zero certificate may reflect insufficient sensitivity, as the amplitude-damping example illustrates. For experiments with joint record data, direct estimation can be preferable; the present inference is suited to separate calibrations alongside retained-pair monitoring. Within those limits, it supplies an optimal quantitative constraint that independent-record descriptions must satisfy.

Appendix A. Sequential preparation of the attaining states

The construction (25) can be generated with a count memory M of dimension $R+1$. Prepare it conditionally on A in

$$|m_s\rangle = \sum_{k=0}^{R} \sqrt{r_k^{(s)}}|k\rangle, \qquad r_k^{(0)} = r_k, \quad r_k^{(1)} = r_{R-k}. \tag{A1}$$

With n output slots remaining, emit a fresh pair F_j, G_j through

$$|k\rangle_M|00\rangle \longmapsto \sqrt{\frac{n-k}{n}}|k\rangle_M|00\rangle + \sqrt{\frac{k}{n}}|k-1\rangle_M|11\rangle, \quad 0 \le k \le n. \tag{A2}$$

At $k=0$, the second term is omitted. The images of distinct reachable count states are orthogonal, so the map has a unitary extension. After R emissions the memory is zero. A particular string of initial weight k has squared amplitude $r_k^{(s)}k!(R-k)!/R! = r_k^{(s)}/\binom{R}{k}$, proving (25).

The memory requires $\lceil \log_2(R+1) \rceil$ qubits. Controlled rotations, conditional decrement and CNOT copying give a polynomial-size implementation. Once the conditional memory preparation is complete, emissions act only on the environment and preserve the reduced AB state. This construction establishes state preparation and distribution, not a memoryless local semigroup realization of every extremizer.

Appendix B. A general affinity bound and numerical upper bound

In the equal-error case, let $Q_s = \prod_j P_s^j$ and $K_s = D(P_s\|Q_s)$. The square-root vectors satisfy

$$\|\sqrt{P_s} - \sqrt{Q_s}\|_2^2 = 2[1 - B(P_s, Q_s)] \le K_s. \tag{B1}$$

The last inequality follows from $D(P_s\|Q_s) \ge -2\ln B(P_s, Q_s) \ge 2[1 - B(P_s, Q_s)]$. Expanding the difference of the two affinities and applying Cauchy–Schwarz gives

$$B(P_0, P_1) \le B(Q_0, Q_1) + \sqrt{K_0} + \sqrt{K_1} \le b^R + 2\sqrt{K}. \tag{B2}$$

For bounded asymmetric errors, apply the local processing in (21) first. Equation (B2) gives the elementary lower bound $K \ge [c - b^R]_+^2/4$ for a required overlap c.

A stronger comparison follows from the angle between square-root probability vectors. Define $\vartheta(P, Q) = \arccos B(P, Q)$. This is their spherical distance and satisfies the triangle inequality. The KL bound used in (B1) implies $\vartheta(P_s, Q_s) \le \phi(K_s)$, where $\phi(x) = \arccos e^{-x/2}$ is concave for $x \ge 0$. Since $B(P_0, P_1) \ge c$ and $K = (K_0 + K_1)/2$, the triangle inequality and Jensen's inequality give

$$\arccos(b^R) \le \vartheta(P_0, Q_0) + \vartheta(P_0, P_1) + \vartheta(P_1, Q_1) \le 2\phi(K) + \arccos c.$$

Inverting this relation yields the comparison plotted in Figure 1,

$$K \geq K_{\mathrm{ang}}(R,\delta,c) := -2\ln\cos\left(\frac{[\arccos(b^R)-\arccos c]_+}{2}\right), \quad 0 \leq c \leq b, \tag{B3}$$

where $[x]_+ = \max\{x,0\}$. The local processing extends it to the original bounded-error distributions because it increases affinity and cannot increase K. This general geometric estimate does not impose all the marginal constraints used by the sharp optimization. At $R = 2, \delta = 0.05, S = 2.10$, it gives approximately 0.00454 nats, compared with the sharp minimum of approximately 0.02540 nats.

A numerical upper bound for (8) follows from positive weights $w_k w_{R-k} = 1$, with $w_{R/2} = 1$ when R is even. Weighted AM–GM gives $\sum_k \sqrt{r_k r_{R-k}} \leq \sum_k w_k r_k$. The relative-entropy variational inequality then yields, for every $\lambda > 0, \beta \in \mathbb{R}$,

$$V_{R,\delta}(\kappa) \leq \lambda\kappa + \beta R\delta + \lambda \ln \sum_k q_k \exp\left(\frac{w_k - \beta k}{\lambda}\right). \tag{B4}$$

At a positive interior optimizer, use $w_k = \sqrt{r_{R-k}/r_k}$ and determine the multipliers from

$$w_k - \lambda[\ln(r_k/q_k) + 1] - \alpha - \beta k = 0. \tag{B5}$$

Equation (B4) remains an upper bound for any such positive reciprocal weights and any $\lambda > 0, \beta$, whether or not numerical stationarity is exact. The computations compare this bound with a feasible primal objective and report residuals separately.

Appendix C. Imbalance, leakage and amplitude damping

If the state remains in $\mathcal{H}_{\mathrm{mc}}$ but has populations $\pi, 1-\pi$, with $0 < \pi < 1$, purification gives branch amplitudes $\sqrt{\pi}, \sqrt{1-\pi}$. The same record argument then implies

$$S_{\max} \leq 2\sqrt{1 + 4\pi(1-\pi)V_{R,\delta}(\kappa)^2}. \tag{C1}$$

Here K remains the unweighted average of the two branch-conditional total correlations in (5). For unequal populations it is not the probability-weighted conditional total correlation. Zero-population endpoints are nonviolating limits, at which one conditional branch is undefined.

For arbitrary leakage, let

$$\Pi = |00\rangle\langle 00| + |11\rangle\langle 11|, \quad \eta = 1 - \mathrm{Tr}(\Pi\rho_{AB}), \quad \rho' = \frac{(\Pi\otimes I)\rho(\Pi\otimes I)}{1-\eta}, \quad 0 \leq \eta < 1. \tag{C2}$$

If the errors and dependence are calibrated within ρ', and its branch populations are $\pi, 1-\pi$, normalized gentle projection gives $\|\rho - \rho'\|_1 \leq 2\sqrt{\eta}$. Since a CHSH operator has norm at most $2\sqrt{2}$,

$$S_{\max}(\rho_{AB}) \le \min\Big\{2\sqrt{2},\, 2\sqrt{1+4\pi(1-\pi)V_{R,\delta}(\kappa)^2}+4\sqrt{2}\sqrt{\eta}\Big\}. \tag{C3}$$

This is a conservative general correction. The Bell value is measured on the unconditioned pair; the projected-sector calibration is obtained on separate runs. Substitution of unconditioned record errors for those conditional errors would not justify (C3).

A sharper formula holds for the X-state subclass with $\rho_{01,10} = 0$. Put $\eta = \rho_{01,01}+\rho_{10,10}$, $\pi = \rho_{00,00}/(1-\eta)$, and $z = \rho_{00,11}$. The correlation-tensor singular values are $2|z|, 2|z|, d$, where $d = |1-2\eta|$. As $2|z| \le a := 2(1-\eta)\sqrt{\pi(1-\pi)}V_{R,\delta}(\kappa)$,

$$S_{\max} \le 2\sqrt{a^2+\max\{a^2,d^2\}}. \tag{C4}$$

The bound is sharp in this subclass: add odd-sector populations with orthogonal unobserved flags to the attaining even-sector construction. For $y = S^2/4$, inversion requires

$$4|z|^2 \ge \max\{0, \min(y-d^2, y/2)\}. \tag{C5}$$

Dividing by $4(1-\eta)^2\pi(1-\pi)$, when nonzero, gives the required squared overlap for (14). If η, π are uncertain, optimize over their joint confidence region; the right-hand side of (C4) need not increase with η.

Independent amplitude damping of Φ^+, with survival probabilities $q_j = e^{-\gamma_j t}$, has this form:

$$\eta = \frac{q_A+q_B-2q_Aq_B}{2}, \qquad z = \frac{\sqrt{q_Aq_B}}{2}. \tag{C6}$$

Thus the record bound can be applied to this relaxation family when its projected-sector record calibrations satisfy (4). The reduced-state form alone does not supply those calibrations.

For the natural two-reservoir dilation, this extension has a specific sensitivity limit. Put $r = (1-q_A)(1-q_B)$, taking finite positive damping times and positive rates; endpoint values below are understood as limits. The normalized even-sector environment branches are $(|00\rangle+\sqrt{r}|11\rangle)/\sqrt{1+r}$ and $|00\rangle$, with overlap $c = 1/\sqrt{1+r}$. Each local reservoir has minimum worst-branch readout error $\delta_* = 1/(2+r)$: report pointer zero upon an emission, and report zero with probability δ_* upon vacuum. This equalizes the two conditional errors; the conditional local states commute, so no other local POVM improves their maximum. Yet

$$c \le 4\delta_*(1-\delta_*) = \frac{4(1+r)}{(2+r)^2}, \qquad 0 \le r \le 1,$$

because $16(1+r)^3-(2+r)^4 = r(16+24r+8r^2-r^3) \ge 0$. Thus (14) yields zero for this natural reservoir realization, although its conditional readouts can be dependent. The shared-memory process in Section 5 supplies a positive certification example. This limitation prevents the structural amplitude-damping extension from being mistaken for a guarantee of detection.

Appendix D. Local and parity monitoring with matched information curves

The theorem assumes records of Z_A, rather than records of an arbitrary observable of the pair. A collective observable can preserve entanglement within its eigenspaces, as is already recognized in composite-system and algebraic accounts of objectivity [22, 23]. To delimit the local-record assumption, compare

$$\mathcal{L}_Q(\rho) = \gamma(Q\rho Q - \rho), \qquad Q_L = Z_A, \quad Q_P = Z_A Z_B. \tag{D1}$$

Let $p = (1 + 1/\sqrt{2})/2$, and initialize the local model in $\sqrt{p}|00\rangle + \sqrt{1-p}|11\rangle$, and the parity model in $\sqrt{p}|\Phi^+\rangle + \sqrt{1-p}|\Psi^+\rangle$. Both start at $S_{\max} = \sqrt{6}$. Their generators have the same eigenvalues and multiplicities, and the two states have equal entropy at every time. Under corresponding pure monitoring dilations, their fragment information curves also agree. Their internal Bell dynamics are nevertheless

$$S_L(t) = 2\sqrt{1 + \tfrac{1}{2}e^{-4\gamma t}}, \qquad S_P(t) = \sqrt{6}. \tag{D2}$$

The parity record describes a relation between the qubits, leaving Bell-state coherence resources inside its sectors. It does not provide the local pointer information required by (4). The equal information curves follow from the two-state structure of each monitoring problem, as shown below.

For either generator in (D1), write the initial state as $\sqrt{p}|u_0\rangle + \sqrt{1-p}|u_1\rangle$, where $Q|u_s\rangle = (-1)^s|u_s\rangle$. The off-diagonal element in this sector basis decays as $e^{-2\gamma t}$. The two nonzero eigenvalues of the reduced state are therefore

$$\lambda_\pm(t) = \frac{1 \pm \sqrt{(2p-1)^2 + 4p(1-p)e^{-4\gamma t}}}{2}. \tag{D3}$$

They are identical for local and parity monitoring. The two Liouvillians also have eigenvalues zero and -2γ, each with multiplicity eight on the two-qubit operator space, because both observables have two-dimensional positive and negative eigenspaces.

For the local initial state, the correlation tensor is

$$\mathrm{diag}(2\sqrt{p(1-p)}e^{-2\gamma t}, -2\sqrt{p(1-p)}e^{-2\gamma t}, 1).$$

For the real Bell-state superposition in the parity model it is

$$\mathrm{diag}(1, -(2p-1), 2p-1).$$

Taking $2p - 1 = 1/\sqrt{2}$ proves (D2).

In homogeneous pure monitoring dilations, a fragment accounting for a fraction f of the monitoring rate has conditional branch overlap $e^{-2\gamma f t}$. Its Holevo information about the sector label is

$$\chi(Q:F) = h_2\left(\frac{1 + \sqrt{(2p-1)^2 + 4p(1-p)e^{-4\gamma f t}}}{2}\right), \tag{D4}$$

where h_2 is binary entropy in bits. The fragment information and any redundancy criterion based on these same curves agree between the models. This comparison concerns matched conditional branch information; it does not identify the global pure branching state with an already decohered spectrum-broadcast mixture.

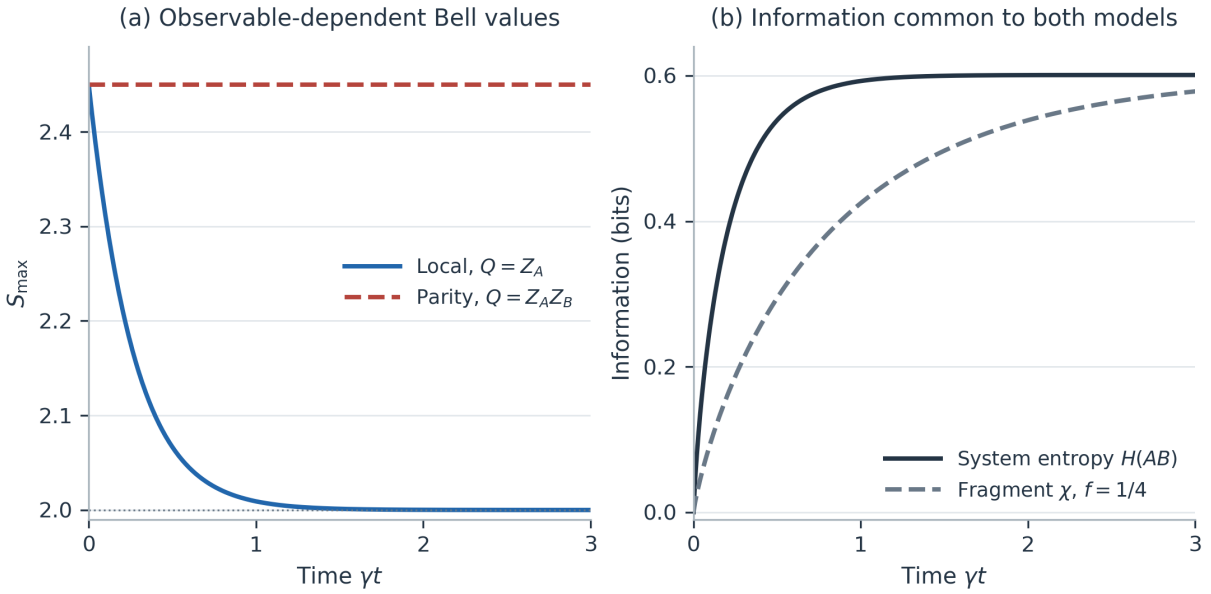


Figure D1. The calibration must identify the recorded observable. Local monitoring uses $Q = Z_A$ with initial state $\sqrt{p}|00\rangle + \sqrt{1-p}|11\rangle$; parity monitoring uses $Q = Z_A Z_B$ with initial state $\sqrt{p}|\Phi^+\rangle + \sqrt{1-p}|\Psi^+\rangle$, where $p = (1 + 1/\sqrt{2})/2$. (a) Both begin at $S_{\max} = \sqrt{6}$, but local monitoring reduces the Bell value while parity monitoring preserves it. (b) Both models have the same system entropy $H(AB)$ and the same pointer Holevo information $\chi(Q : F)$ in a homogeneous bath fragment carrying the rate fraction $f = 1/4$. These common curves are in bits. The comparison concerns different recorded observables and different initial states, with matched initial Bell values, generator spectra and entropy trajectories.

Data and code availability

All reported data are generated by the stated models. The accompanying **Computational Supplement** provides executable Python source, parameter choices, numerical diagnostics and figure-generation instructions in Markdown. The archive also includes the source files, generated data and figures; `README.md` gives the reproduction commands. No experimental dataset is used.

Declaration of AI assistance

AI was used in mathematical exploration, literature discovery, code development and manuscript preparation. The author is responsible for the content and for the final version submitted for publication.

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